EPM213s - Electrical Measurements

Dynamometer Inspection and Construction

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Introduction

A dynamometer is a device designed to evaluate quantities such as force, torque, and power. Within the field of electrical engineering, it is commonly employed to measure electrical power in AC circuits, often through electrodynamometer-based wattmeters.

Classifications of Dynamometers

- Absorption dynamometers

- Transmission dynamometers

- Electrodynamometer-type wattmeters (focus of this report)

Electrodynamometer Wattmeter Structure

An electrodynamometer wattmeter consists of two coils:

- Fixed Coil: It is linked in series with the load, allowing it to carry the load current.

- Moving Coil: It is connected across the voltage supply using a series resistor.

Other parts include a spring mechanism for torque restoration and a damping setup, generally utilizing air friction or eddy currents.

Diagram

Materials Used and Design Considerations

- Coil: Copper wire (assumed due to high conductivity)

- Frame: Non-magnetic, insulating material

- Damping: Air friction type (common in labs)

Inspection Criteria

- Check for mechanical wear and misalignment.

- Ensure electrical continuity in both fixed and moving coils.

- Confirm calibration against known standards.

- Verify free movement of moving coil.